

Problem 1:

Suppose that $G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ is a secure pseudorandom generator that expands an n -bit random seed into a $2n$ -bit pseudorandom string. For each of the following questions: first, give a YES/NO answer; if the answer is YES, please prove it; if the answer is NO, please provide a counter-example

(a)

Let $\wedge$ denote the bit-wise and operation. Is $G((0^{n-1}||1) \wedge s)$ a secure pseudorandom generator that expands a random seed of length n into a pseudorandom string of length $2n$?

Proof.

No, this is not secure since for $G(0^{n-1}||1 \wedge s)$ for a random n -bit string s . $G(0^{n-1}||1 \wedge s)$ is either $G(0^n)$ or $G(0^{n-1}||1)$.

Now define a distinguisher D s.t.

$$D(y) = \begin{cases} 1 & \text{if } y \in \{G(0^n), G(0^{n-1}||1)\}, \\ 0 & \text{otherwise.} \end{cases}$$

If $y \leftarrow G((0^{n-1}||1) \wedge U_n)$, then $\mathbb{P}[D(y) = 1] = 1$

If $y \leftarrow U_{2n}$, then $\mathbb{P}[D(y) = 1] = 2/2^{2n}$

Thus,

$$\mathbb{P}[D(G(0^{n-1}||1) \wedge U_n) = 1] - \mathbb{P}[D(U_{2n}) = 1] \geq 1 - 2^{1-2n}$$

this is not negligible. Therefore, this is not a secure PRG. ■

(b)

Is $G(\text{first half of } G(s)) || G(\text{second half of } G(s))$ a secure pseudorandom generator that maps n bits to $4n$ bits?

Proof.

Consider these distributions:

- $X_0 = \{G(\text{first half of } G(s)) || G(\text{second half of } G(s)) \mid s \xleftarrow{\$} \{0, 1\}^n\}$
- $X_1 = \{G(\text{first half of } s) || G(\text{second half of } s) \mid s \xleftarrow{\$} \{0, 1\}^{2n}\}$

$$= \{G(x)||G(y) \mid x, y \stackrel{\$}{\leftarrow} \{0, 1\}^n\}$$

- $X_2 = \{G(s)||x \mid s \stackrel{\$}{\leftarrow} \{0, 1\}^n \text{ and } x \stackrel{\$}{\leftarrow} \{0, 1\}^{2n}\}$
- $X_3 = \{x||y \mid x, y \stackrel{\$}{\leftarrow} \{0, 1\}^{2n}\} = \{s \mid s \stackrel{\$}{\leftarrow} \{0, 1\}^{4n}\}$

Since we know that G is a secure PRG we know that

$$\{G(s) \mid s \stackrel{\$}{\leftarrow} \{0, 1\}^n\} \approx \{s \mid s \stackrel{\$}{\leftarrow} \{0, 1\}^{2n}\}$$

This implies that $X_0 \approx X_1$ since a $2n$ -bit string from G or uniformly random looks the same, so taking the first and second half will also look the same.

Again since we know that

$$\{G(s) \mid s \stackrel{\$}{\leftarrow} \{0, 1\}^n\} \approx \{s \mid s \stackrel{\$}{\leftarrow} \{0, 1\}^{2n}\}$$

This implies that $X_1 \approx X_2$, then again use the same thing, and we get $X_2 \approx X_3$.

Therefore, by hybrid argument we have shown that $X_0 \approx X_3$, and thus, this is a secure PRG. ■

Problem 2:

Determine whether the following $F : \{0, 1\}^{2n} \times \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ is a pseudo-random function. If yes, prove it; if not, describe and analyze a feasible distinguisher that has non-negligible advantage in distinguishing F from random function family.

$$F_k(x) := G(x) \oplus k \text{ where } G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n} \text{ is a PRG.}$$

Proof.

No, this is not a secure PRF.

First choose distinct $x_1, x_2 \in \{0, 1\}^n$. Query the oracle twice to get $y_i \leftarrow O(x_i)$ then define D s.t.

$$D(y_1, y_2) = \begin{cases} 1 & \text{if } y_1 \oplus y_2 = G(x_1) \oplus G(x_2), \\ 0 & \text{otherwise.} \end{cases}$$

Thus, $\mathbb{P}[D(y_1, y_2) = 1 \mid y_i = G(x_i)] = 1$, but for a truly random function F , since y_1, y_2 are independent and uniform across $\{0, 1\}^{2n}$ this implies that $y_1 \oplus y_2$ is also uniform across $\{0, 1\}^{2n}$. Therefore, $\mathbb{P}[D(y_1, y_2) = 1 \mid y_i = F(x_i)] = 2^{-2n}$.

Therefore, the advantage is $1 - 2^{-2n}$ which is non negligible. ■

Problem 3:

Give a distinguisher that has non-negligible advantage in distinguishing the following $F' : \{0, 1\}^{2n} \times \{0, 1\}^{2n} \rightarrow \{0, 1\}^{2n}$ from a random function family.

$F'_{k_1||k_2}(x_1||x_2) := F_{k_1}(x_1)||F_{k_2}(x_2)$ where $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ is a PRF and $|x_1| = |x_2|$. Here k_1 and k_2 are drawn uniformly from the keyspace of F .

Proof.

First choose distinct $x_1, x_2 \in \{0, 1\}^n$. Query the oracle three times

- $O(x_1||x_1) = y_1||y'_1$
- $O(x_1||x_2) = y_2||y'_2$
- $O(x_2||x_1) = y_3||y'_3$

then define D s.t.

$$D(y_1||y'_1, y_2||y'_2, y_3||y'_3) \begin{cases} 1 & \text{if } y_1 = y_2 \text{ and } y'_1 = y'_3, \\ 0 & \text{otherwise.} \end{cases}$$

Thus, $\mathbb{P}[D(y_1||y'_1, y_2||y'_2, y_3||y'_3) = 1 \mid \mathcal{O} = F'] = 1$, but for a truly random function H , since $x_1||x_1, x_1||x_2, x_2||x_1$ are all distinct this implies that the three outputs from H are all independent and uniform over $\{0, 1\}^{2n}$. Thus, $\mathbb{P}[D(y_1||y'_1, y_2||y'_2, y_3||y'_3) \mid \mathcal{O} = H] = 2^{-n} \cdot 2^{-n} = 2^{-2n}$.

Therefore, the advantage is $1 - 2^{-2n}$ which is non negligible. ■

Problem 4:

Consider two encryption schemes $\Pi_1 = (\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ and $\Pi_2 = (\text{Gen}_2, \text{Enc}_2, \text{Dec}_2)$ over the key space $\mathcal{K}$, message space $\mathcal{M}$ and ciphertext $\mathcal{C}$. Suppose that $\mathcal{K} = \mathcal{M} = \mathcal{C}$. Now suppose that at least one of Π_1 and Π_2 is CPA secure but you do not know which one. Construct a CPA secure encryption scheme using Π_1 and Π_2 .

Please describe your construction and prove it correct and CPA secure, assuming that both encryption scheme are correct, and at least one of the underlying encryption schemes is CPA secure. Besides the two encryption schemes Π_1 and Π_2 , please do not use other cryptographic primitives in your construction.

Proof.

Define $\Pi' = (\text{Gen}', \text{Enc}', \text{Dec}')$ s.t. $\text{Gen}' = \text{Gen}_1 || \text{Gen}_2$, $\text{Enc}'(k_1 || k_2, m) = \text{Enc}_1(k_1, \text{Enc}_2(k_2, m))$, and $\text{Dec}'(k_1 || k_2, c) = \text{Dec}_2(k_2, \text{Dec}_1(k_1, c))$, where $k_1 || k_2$ is the key given from Gen' .

Correctness:

Given $\text{Gen}' = k_1 || k_2$. Let $m \in \mathcal{M}$.

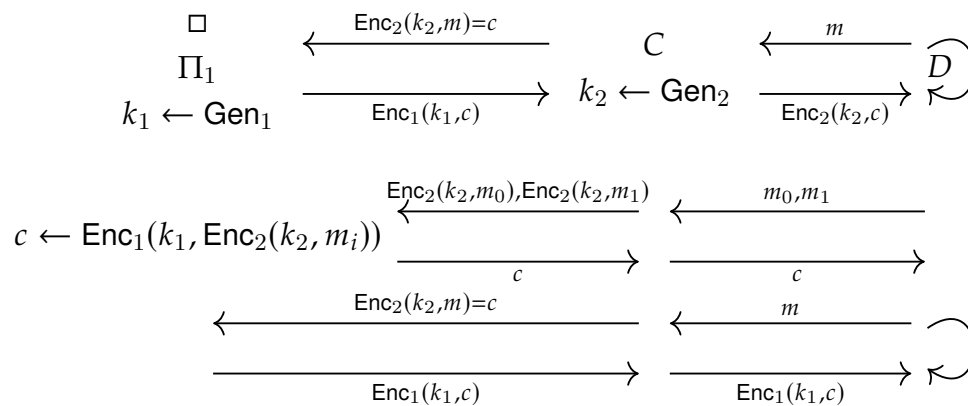
$\text{Dec}'(k_1 || k_2, \text{Enc}'(k_1 || k_2, m)) =$

$\text{Dec}_2(k_2, \text{Dec}_1(k_1, \text{Enc}_1(k_1, \text{Enc}_2(k_2, m)))) = \text{Dec}_2(k_2, \text{Enc}_2(k_2, m)) = m$.

CPA secure:

Case 1: Π_1 is CPA secure.

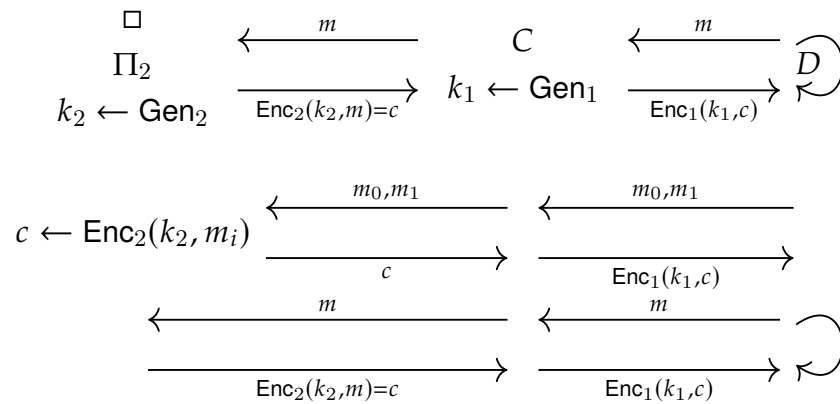
Assume D has non negligible advantage against Π' . Define C s.t.



Then output what D outputs. This implies that C distinguishes Π_2 . This implies if Π_2 is secure then Π' is secure.

Case 2: Π_2 is CPA secure.

Assume D has non negligible advantage against Π' . Define C s.t.



Then output what D outputs. This implies that C distinguishes Π_1 . This implies if Π_1 is secure then Π' is secure.

Therefore, in both cases we know that Π' is secure given Π_1 or Π_2 is secure. ■

Problem 5:

Let $F: \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a PRF. Consider the following MAC = (Gen, Mac, Vrfy) given by

- Gen(1^n): outputs a uniformly random key $k \leftarrow \{0, 1\}^n$.
- Mac(k, m): given a message $m = m_1 || m_2 \in \{0, 1\}^{2n}$, where $|m_1| = |m_2| = n$, output the tag $t := F_k(m_1) || F_k(F_k(m_2))$.
- Vrfy(k, m, t): check whether Mac(k, m) = t .

Is this MAC secure (existential unforgeability under chosen message attack)? Prove your answer either by giving a formal reduction to F being a PRF or by constructing a forger with non-negligible advantage in the security game against MAC.

Proof.

This is not secure. First pick $m \in \{0, 1\}^{2n}$ s.t. $m = m_1 || m_2$ where m_1, m_2 are distinct n -bit strings. Query the oracle two times

- MAC($m_1 || m_1$) = $F_k(m_1) || F_k(F_k(m_1)) = t_1$
- MAC($m_2 || m_2$) = $F_k(m_2) || F_k(F_k(m_2)) = t_2$

then return $(m, F_k(m_1) || F_k(F_k(m_2)))$ we can make this tag since we know the values from the queries before, and since m has never queried before and the tag sent is valid this implies this is not MAC secure. ■

Problem 6:

For each of the following hash functions H , determine whether it is *necessarily* collision resistant. If it is, describe and analyze an appropriate reduction. If it's not, describe and analyze a feasible attacker against the collision resistance of H .

(a)

$H^s(x) = H_1^s(x) \oplus H_2^s(x)$, where $\{H_1^s\}, \{H_2^s\}$ are both collision-resistant hash function families.

Proof.

False, let $\{H_1^s\}$ be a collision-resistant hash function family and let $\{H_2^s\} = \{H_1^s\}$. These are both collision resistant

Then consider $H^s(x) = H_1^s(x) \oplus H_2^s(x) = H_1^s(x) \oplus H_1^s(x) = 0^n$.

Thus, define an attacker D s.t. D returns $x, y \in \{0, 1\}^n$ s.t. $x \neq y$. Then $H^s(x) = 0^n = H^s(y)$. Thus, with probability 1 D can find a collision.

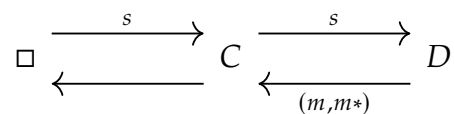
Therefore, H is not necessarily collision resistant. ■

(b)

$H^s(x) = \tilde{H}^s(\tilde{H}^s(x))$, where $\{\tilde{H}^s\}$ is a collision-resistant hash function family (that can support arbitrary length inputs).

Proof.

Assume there exists an adversary D that can produce a valid collision for H^s with non negligible probability.



Case 1: $\tilde{H}^s(m) = \tilde{H}^s(m^*)$ return m, m^*

Case 2: $\tilde{H}^s(m) \neq \tilde{H}^s(m^*)$ return $\tilde{H}^s(m), \tilde{H}^s(m^*)$

Therefore, in both cases we provide a valid collision for $\tilde{H}^s$ with non negligible probability.

Therefore, this implies if $\{\tilde{H}^s\}$ is a collision-resistant hash function family then $\{H^s\}$ is also. ■